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A Long Version of a Short Paper

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Abstract

Lorem ipsum way too long to be written as a short version anyway. Still, when considering the facts from first principles, we find that the outcomes of this introspective approach is compatible with the guidelines previously established.

In such an experiment it is then clear that the potential for further development not only depends on previous relationships found but also on connections made during exploitation of this novel new experimental protocol.

Keywords terraforming, desert, numerical perspective

1. INTRODUCTION

Twelve hundred years ago—in a galaxy just across the hill ...

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2. BIBLIOGRAPHIES, CITATIONS AND BLOCK QUOTES

We encourage you to include a `.bib` file, which will be automatically included by MyST.

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To reference citations contained in that bibliography use the `\cite` command, as in [1] (which literally is `\cite{hume48}` in accordance with the `hume48` cite-key in the associated `mybib.bib` file).

If you wish to have a block quote, use the `\begin{quote}` command, as in:

When it is asked, What is the nature of all our reasonings concerning matter of fact? the proper answer seems to be, that they are founded on the relation of cause and effect. When again it is asked, What is the foundation of all our reasonings and conclusions concerning that relation? it may be replied in one word, experience. But if we still carry on our sifting humor, and ask, What is the foundation of all conclusions from experience? this implies a new question, which may be of more difficult solution and explication. D. Hume [1]

== DOIs in bibliographies

In order to include a DOI in your bibliography, add the DOI to your bibliography entry as a string. For example:

```
@book{hume48,
  author   = {David Hume},
  year     = {1748},
  title    = {An enquiry concerning human understanding},
  address  = {Indianapolis, IN},
  publisher = {Hackett},
  doi      = {10.1017/CB09780511808432}
}
```

3. CITING SOFTWARE AND WEBSITES

Any paper relying on open-source software would surely want to include citations. Often you can find a citation in BibTeX format via a web search. Authors of software packages may even publish guidelines on how to cite their work.

For convenience, citations to common packages such as Jupyter [2], Matplotlib [3], NumPy [4], pandas [5], [6], scikit-learn [7], [8], and SciPy [9] are included in this paper's `.bib` file.

In this paper we not only terraform a desert using the package `terradesert` [10], we also catch a sandworm with it. To cite a website, the following BibTeX format plus any additional tags necessary for specifying the referenced content is recommended. If you are citing a team, ensure that the author name is wrapped in additional braces `{Team Name}`, so it is not treated as an author's first and last names.

```
@misc{terradesert,
  author = {{TerraDesert Team}},
  title  = {Code for terraforming a desert},
  year   = {2000},
  url    = {https://terradesert.com/code/},
  note   = {Accessed 1 Jan. 2000}
}
```

4. SOURCE CODE EXAMPLES

Of course, no paper would be complete without some source code. Without specifying a language the highlighting looks like this:

```
def sum(a, b):
    """Sum two numbers."""

    return a + b
```

Program 1. *Example Python Code*

```
def sum(a, b):
    """Sum two numbers."""

    return a + b
```

The `mystmd` parser recognizes `lstlisting` code-highlighting syntax:

```
def sum(a, b):
    """Sum two numbers."""

    return a + b
```

You can also refer to a code block with a caption [Program 1](#).

Maybe also in another language, and with line numbers:

```
1 int main() {
2     for (int i = 0; i < 10; i++) {
3         /* do something */
4     }
5     return 0;
6 }
```

Or a snippet from the above code, starting at the correct line number:

```
for (int i = 0; i < 10; i++) {
    /* do something */
}
```

The code will be properly formatted and presented with a copy-code button in the HTML view of your paper.

5. IMPORTANT PART

It is well known that Spice grows on the planet Dune [11]. Test some maths, for example $e^{\pi i} + 3\delta$. Or maybe an equation on a separate line:

$$g(x) = \int_0^\infty f(x)dx \quad (1)$$

or using aligned lines for a single equation:

$$\begin{aligned} g(x) &= \int_0^\infty f(x)dx \\ &= \dots \end{aligned} \quad (2)$$

The area of a circle and volume of a sphere are given as

$$A(r) = \pi r^2. \quad (3)$$

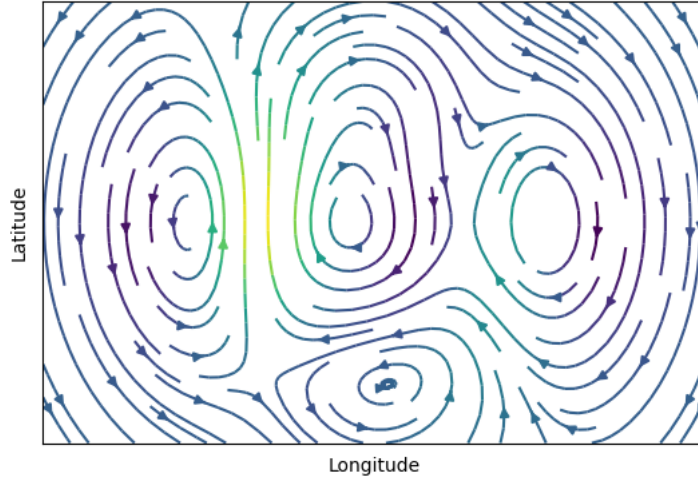


Figure 1. This is the caption, sandworm vorticity based on storm location in a pleasing stream plot. Based on example in [matplotlib](#).

$$V(r) = \frac{4}{3}\pi r^3 \quad (4)$$

we can then refer back to Equation (2) or (3) later.

Mauris purus enim, volutpat non dapibus et, gravida sit amet sapien. In at consectetur lacus. Praesent orci nulla, blandit eu egestas nec, facilisis vel lacus. Fusce non ante vitae justo faucibus facilisis. Nam venenatis lacinia turpis. Donec eu ultrices mauris. Ut pulvinar viverra rhoncus. Vivamus adipiscing faucibus ligula, in porta orci vehicula in. Suspendisse quis augue arcu, sit amet accumsan diam. Vestibulum lacinia luctus dui. Aliquam odio arcu, faucibus non laoreet ac, condimentum eu quam. Quisque et nunc non diam consequat iaculis ut quis leo. Integer suscipit accumsan ligula. Sed nec eros a orci aliquam dictum sed ac felis. Suspendisse sit amet dui ut ligula iaculis sollicitudin vel id velit. Pellentesque hendrerit sapien ac ante facilisis lacinia. Nunc sit amet sem sem. In tellus metus, elementum vitae tincidunt ac, volutpat sit amet mauris. Maecenas¹ diam turpis, placerat² at adipiscing ac, pulvinar id metus.

As you can see in [Figure 1](#), the vorticity is high in the margins. Additionally in Figures [1](#) and [2](#) ... this is how you reference auto-numbered figures. Use `autoref` when referring to a single figure, and `ref` if you need individual access to the number.

We show the different quantities of materials required in [Table 1](#).

And since you are working in raw LaTeX already, you can easily make complex table formats. Above we used `\Dtablewidth` to control the width of individual parts of a table, but you can also let tex worry about that part for you:

Perhaps we want to end off with a quote by Lao Tse³:

Muddy water, let stand, becomes clear.

¹On the one hand, a footnote.

²On the other hand, another footnote.

³ $e^{-i\pi}$

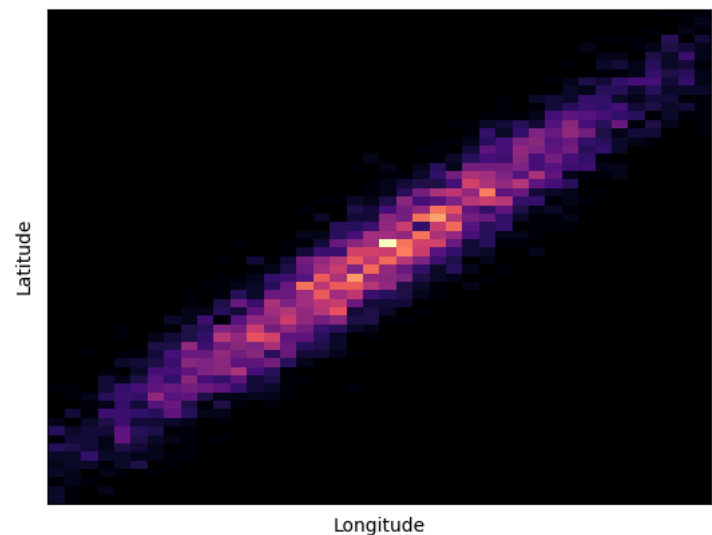


Figure 2. *This is the caption, electromagnetic signature of the sandworm based on remote sensing techniques. Based on example in [matplotlib](#).*

Table 1. *This is the caption for the materials table.*

Material	Units
Stone	3
Water	12
Cement	α

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Table 2. *This is the caption for the wide table with custom widths.*

This	is	a	very	very	wide	table
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Table 3. *Area Comparisons*

Projection	Area in square miles		
	Large Horizontal Area	Large Vertical Area	Smaller Square Area
Albers Equal Area	7,498.7	10,847.3	35.8
Web Mercator	13,410.0	18,271.4	63.0
Difference	5,911.3	7,424.1	27.2
Percent Difference	44%	41%	43%

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